

电机数据：

FL_calf:-0.0052~-1.4042(缩~申)
FL_thigh:-1.6601~1.0000(后~前)
FL_hip:-0.5522~0.4007（上翻~下翻）

FR_calf:0.0175~1.4368 (缩~申)
FR_thigh:1.7418~-0.8261 (后~前)
FR_hip:0.7582~-0.4049 （上翻~下翻）

RL_calf:-0.0136~-1.3881(缩~申)
RL_thigh:-1.6938~0.0715 (后~前)
RL_hip:0.8384~-0.4061（上翻~下翻）

RR_calf:0.0117~1.4629(缩~申)
RR_thigh:1.7851~-0.0681(后~前)
RR_hip:-0.6692~0.4436（上翻~下翻）

方向（很多：可能是一半的电机）+零点（零点不符合的小腿）

当前拟合函数（Torque-Speed Envelope）

设转速为 n (rpm)，先定义二次拟合函数：（为了保证不超过上限，c的数值从最接近的14.0886978降低为13.605312408）

$$T_q(n) = an^2 + bn + c$$

其中参数为：

$$a = -0.000341191, \quad b = 0.064565966, \quad c = 13.605312408$$

峰值点：

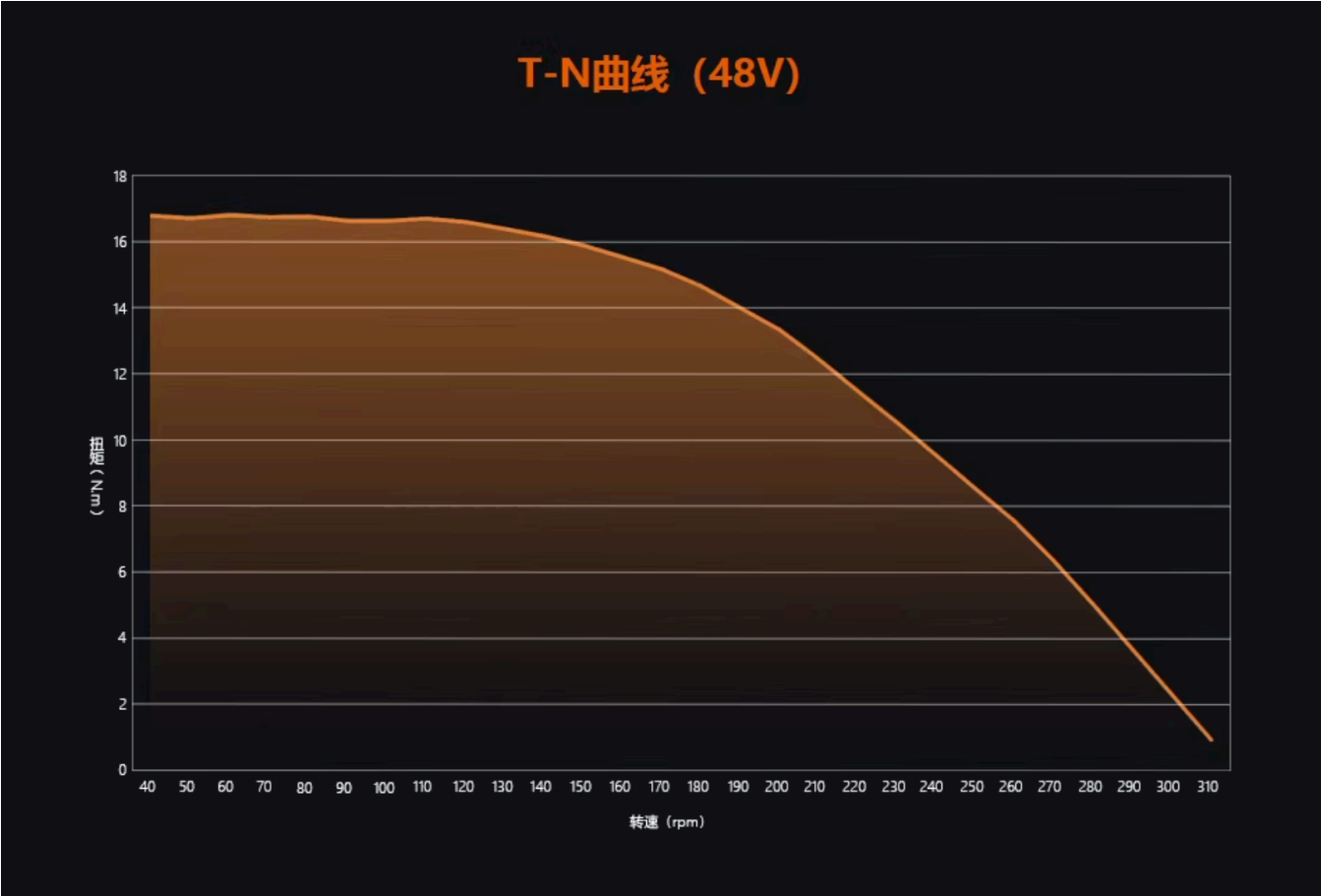
$$n_{\text{peak}} = -\frac{b}{2a} \approx 94.62 \text{ rpm}$$

$$T_{\text{peak}} = T_q(n_{\text{peak}}) \approx 16.6599 \text{ N}\cdot\text{m}$$

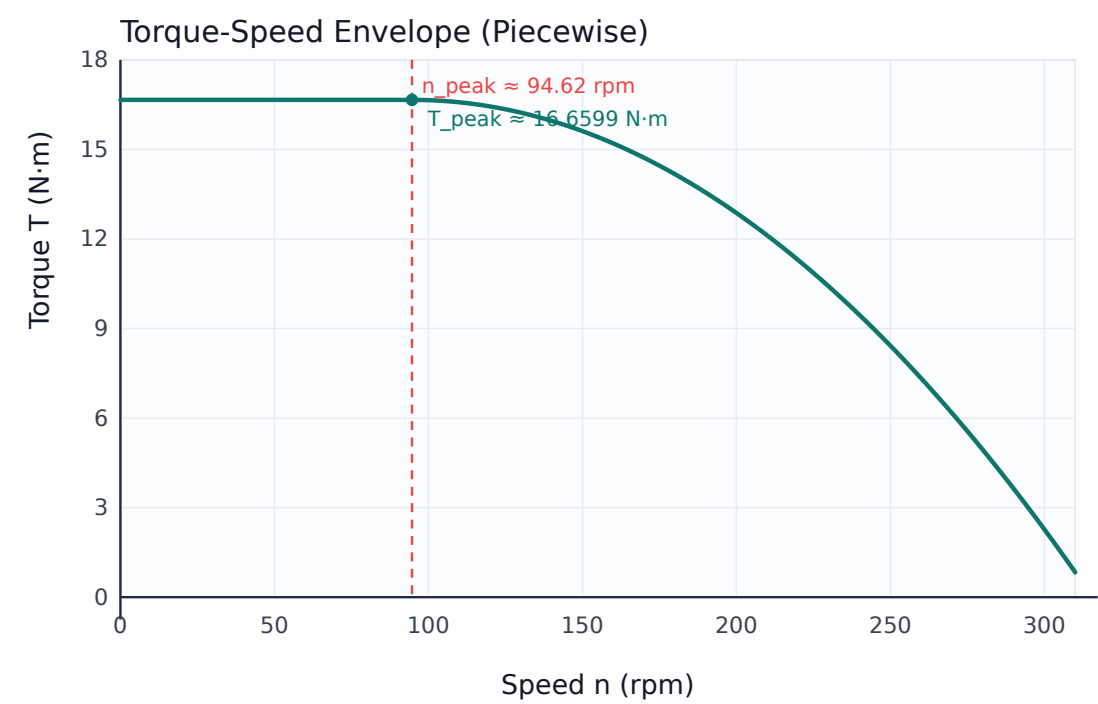
分段包络函数

$$T_{\text{env}}(n) = \begin{cases} T_{\text{peak}}, & n \leq n_{\text{peak}} \\ T_q(n), & n > n_{\text{peak}} \end{cases}$$

函数图像（分段函数）



下图按分段函数绘制，坐标范围固定为： $n \in [0, 310]$ rpm， $T \in [0, 18]$ N·m。()



实现时对正反转对称，使用 $|n|$ 。

物理裁剪后的可用扭矩上限

$$T_{\max}(n) = \min(\max(T_{\text{env}}(|n|), 0), \tau_{\text{sat}}, \tau_{\text{limit}})$$

最终输出扭矩

$$\tau_{\text{applied}} = \text{clip}(\tau_{\text{cmd}}, -T_{\max}, T_{\max})$$

重力补偿数据及其拟合

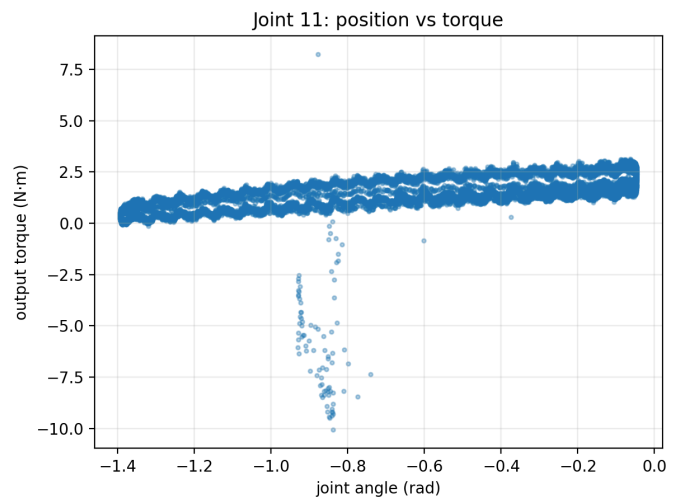
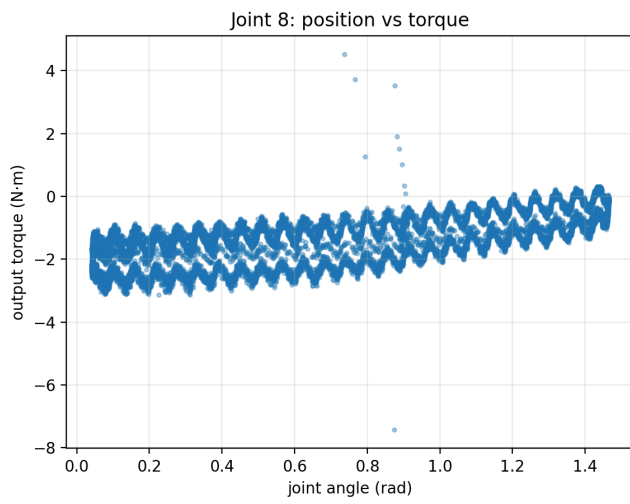
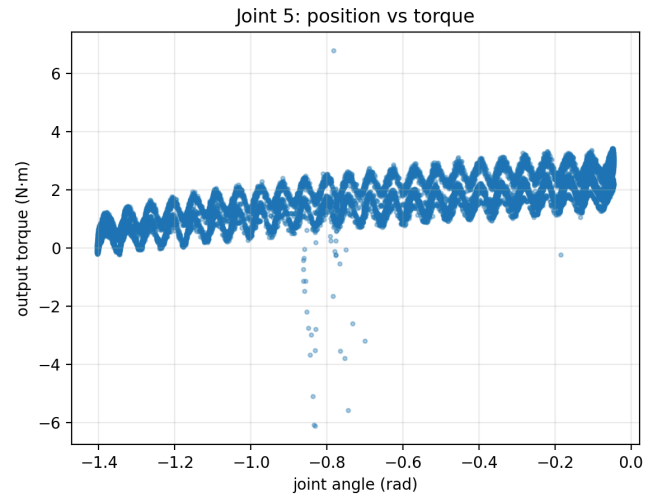
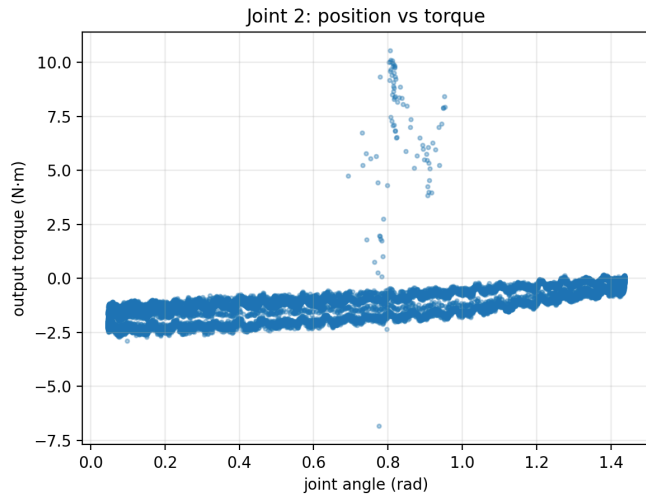
数据获取方式

在小腿电机水平放置的情况下，让小腿电机在之前的限位区间以很小的速度进行摆动360s转动2.8rad，即0.0078rad/s，近似视为每个旋转角度都处于平衡状态。接着在有弹簧的情况和拆卸弹簧的情况下各自测一组数据，最终每组数据大约20000行左右。

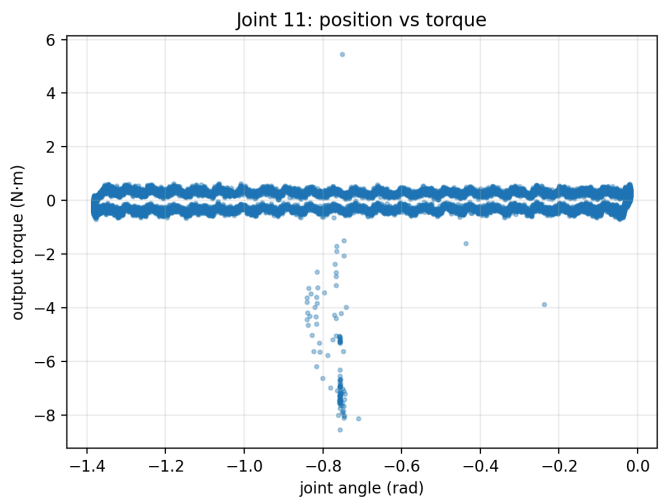
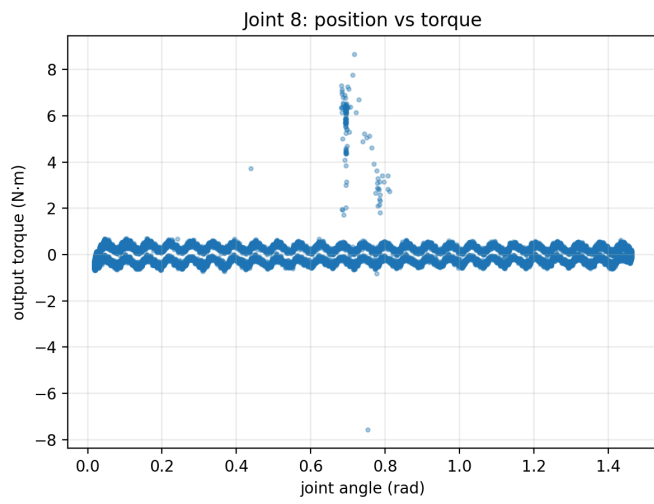
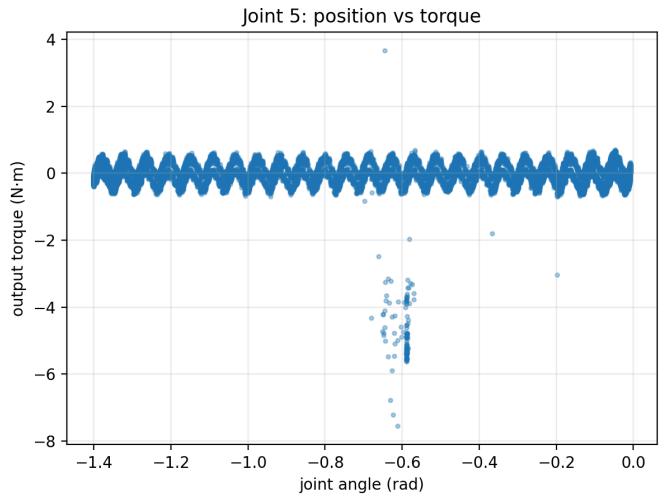
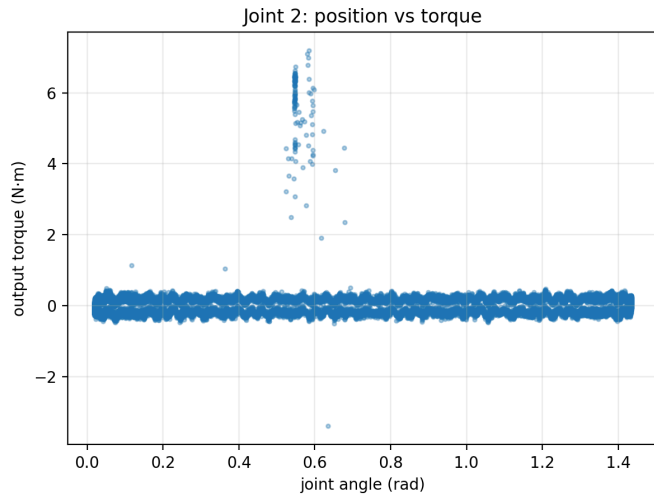
数据处理方式：

- 先做有弹簧/无弹簧的角度插值差分
- 再筛选低速样本，条件是 $|\dot{q}| < 0.05 \text{ rad/s}$
- 再按角度分箱平均，bin 宽度是 0.02 rad
- 最后同时拟合了线性和三角函数

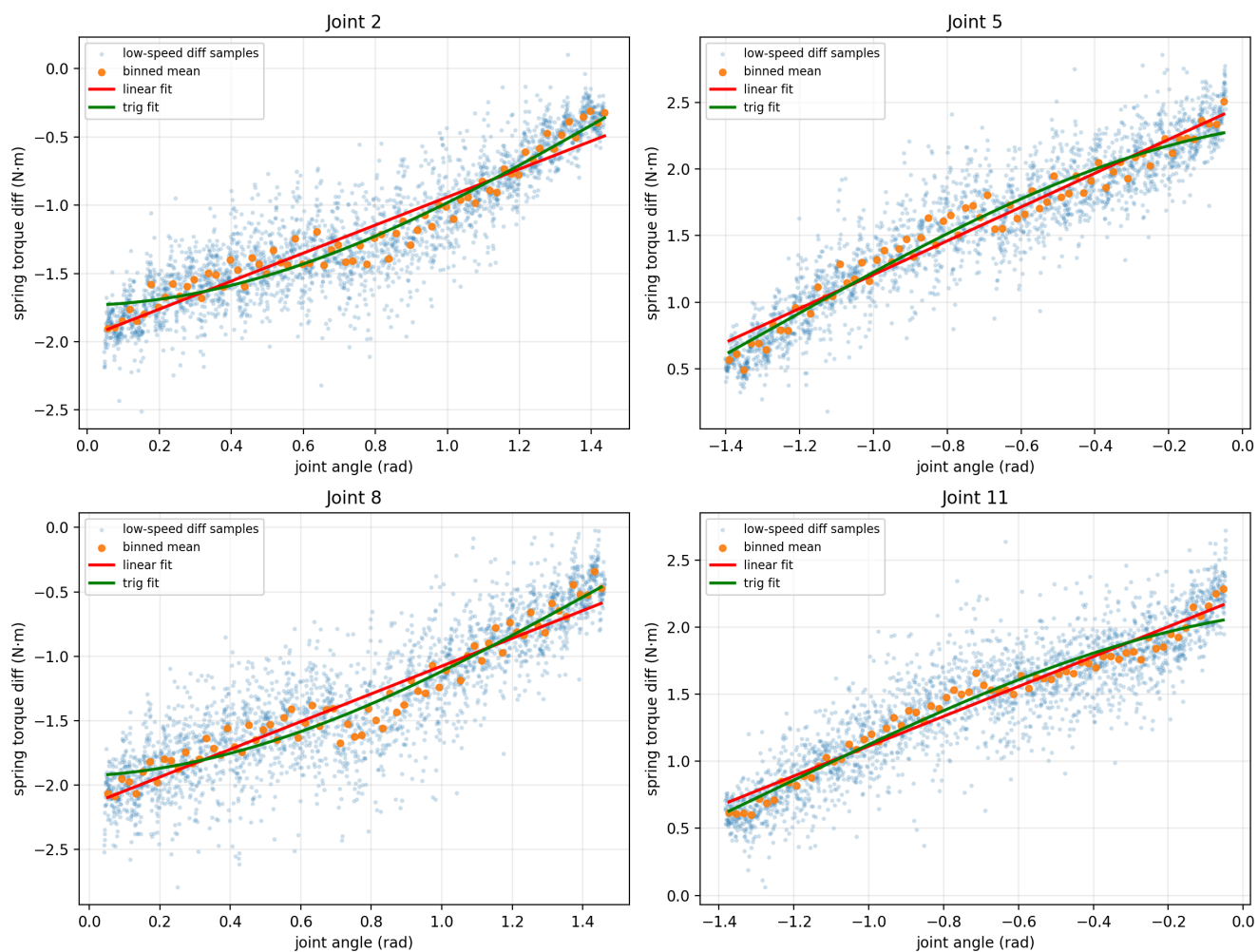
Joint Position - Torque Scatter Plot (motor2.csv)



Joint Position - Torque Scatter Plot (motor3.csv)



Low-speed Binned Spring Torque Difference: motor2 - motor3



线性拟合结果

$$\tau = kq + b \quad \tau = k^*q + b$$

- 关节 2:

$$\tau = 1.026204q - 1.966818, R^2 = 0.9247$$

- 关节 5:

$$\tau = 1.270471q + 2.476247, R^2 = 0.9568$$

- 关节 8:

$$\tau = 1.077613q - 2.153850, R^2 = 0.9257$$

- 关节 11:

$$\tau = 1.113021q + 2.225375, R^2 = 0.9607$$

三角拟合结果

$$\tau = a \sin(q) + b \cos(q) + c$$

- 关节 2:

$$\tau=0.071473\sin(q)-1.504900\cos(q)-0.228728, R^2=0.9492$$

- 关节 5:

$$\tau=0.464990\sin(q)+1.482100\cos(q)+0.814629, R^2=0.9635$$

- 关节 8:

$$\tau=0.145519\sin(q)-1.499135\cos(q)-0.429548, R^2=0.9462$$

- 关节 11:

$$\tau=0.442854\sin(q)+1.262931\cos(q)+0.814852, R^2=0.9655$$